

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 7: Coordinate Geometry

Previous year board questions, topic-wise: 15 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The distance formula

1. The centre of a circle is $(x - 7, 2x)$. The circle passes through $(-9, 11)$ and its radius is $5\sqrt{2}$ units. Find the possible values of x . **CBSE 2026, 2 marks**
2. A circle has its centre at $(2a, a - 7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a . **CBSE 2025, 2 marks**
3. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**
4. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**
5. Find the distance between $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$. (a) $2\sqrt{10}$ units (b) $4\sqrt{10}$ units (c) $2\sqrt{20}$ units (d) 0 **CBSE 2023, 1 mark**
6. Two vertices of an equilateral triangle are $(-5, 3)$ and $(5, 3)$, and the origin lies inside the triangle. Find the third vertex. Take $\sqrt{3} = 1.7$. **CBSE 2023, 3 marks**
7. Assertion (A): $P(0, 2)$ is the point where the line $3x + 2y = 4$ meets the y -axis. Reason (R): The distance of $P(0, 2)$ from the x -axis is 2 units. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2023, 1 mark**

2 The section formula

8. Where does the mid-point of the segment joining $(5, -4)$ and $(6, 4)$ lie? (a) on the x -axis (b) on the y -axis (c) at the origin (d) on neither axis **CBSE 2026, 1 mark**
9. In what ratio does the x -axis divide the segment joining $(-6, 5)$ and $(-4, -1)$? Find the point where they meet. **CBSE 2026, 3 marks**
10. Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? **CBSE 2025, 1 mark**
11. In what ratio does the y -axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. **CBSE 2025, 3 marks**
12. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**
13. The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. **CBSE 2024, 3 marks**

- 14.** $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other.
CBSE 2024, 3 marks
- 15.** In what ratio does the x -axis divide the segment joining $A(3, 6)$ and $B(-12, -3)$? (a) 1 : 2 (b) 1 : 4 (c) 4 : 1 (d) 2 : 1 **CBSE 2023, 1 mark**

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